Choice Modeling Methods and Pairwise Comparisons

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Overview

This document summarizes various methods used to rank or evaluate choices through pairwise comparisons and other preference elicitation techniques.

What Is the Method for Ranking Items by Pairwise Comparisons?

Several methods are commonly used to derive rankings through pairwise comparisons:

1. Elo Rating System

- Used extensively in competitive games (e.g., chess).
- Each item (or player) has a rating.
- When two items are compared, their ratings are updated based on the outcome.
- Over time, these ratings can be used to rank all items.

2. Bradley-Terry Model

- A **probabilistic model** for pairwise comparisons.
- Models the probability that item i is preferred over item j based on their latent "strength" parameters.
- Allows estimation of a ranking or scale.

3. Thurstone's Case V (Paired Comparison Law)

- A psychometric scaling method.
- Assumes normally distributed underlying preferences.
- Differences in preference are modeled to place items on a continuous scale.

4. Rank Centrality

- A Markov chain–based algorithm.
- Uses pairwise comparison probabilities to derive a global ranking.
- Applied in various contexts like recommendation systems and tournaments.

Choice Modeling Approaches

Several of the above methods intersect with **choice modeling**, a broader field focused on modeling how people make selections from sets of alternatives.

1. **Conjoint Analysis**

   - A core technique in choice modeling.
   - Evaluates how people value different attributes of a product or decision.
   - Key types:
     - **Full-profile conjoint**
     - **Choice-based conjoint (CBC)**
     - **Adaptive conjoint analysis (ACA)**

2. **Best-Worst Scaling (BWS)**

   - Participants choose the **most** and **least** preferred items from a set.
   - Generates fine-grained ranking and importance weights.
   - Falls under the umbrella of discrete choice methods.

3. **MaxDiff Scaling**

   - A subtype of Best-Worst Scaling.
   - Particularly useful for ranking large sets of items.
   - Often used in market research and UX studies.

4. **Thurstone's Paired Comparison Model**

   - One of the earliest models in psychometrics for deriving preferences.
   - Serves as a theoretical foundation for modern choice modeling.

5. **Bradley-Terry Model**

   - As mentioned above, this model also fits under the choice modeling umbrella.
   - Useful for estimating comparative preferences when only pairwise comparisons are available.

Are All These Methods Part of Choice Modeling?

Yes — all the methods listed above are part of, or closely related to, the broader field of **choice modeling**. Here's a summary:

- **Included in Choice Modeling**:
  - Conjoint Analysis
  - Best-Worst Scaling (BWS)
  - MaxDiff
  - Bradley-Terry Model
  - Thurstone's Paired Comparison
- **Related But More General**:
  - Elo Rating System
  - Rank Centrality

These methods differ in how they collect and analyze preference data, but they all aim to uncover how people make decisions or assign value to options.

Summary

- Choice modeling is a broad framework that includes methods based on pairwise comparisons and discrete choices.
- Some methods, like **conjoint analysis**, are specifically designed for trade-off analysis among attributes.
- Others, like **MaxDiff** and **Bradley-Terry**, are better suited for ranking or scaling based on direct or indirect comparisons.